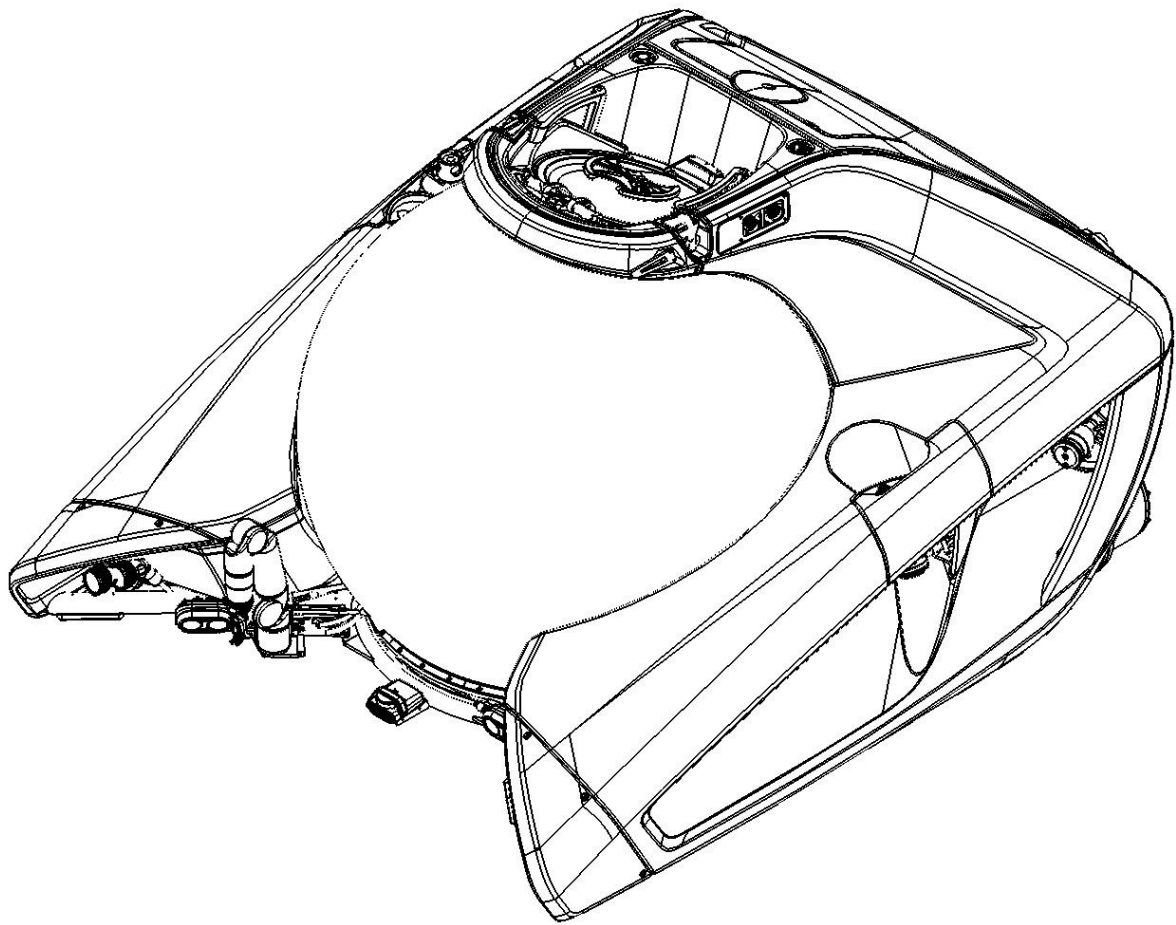


Critical Piloting Numbers

Quick Reference Guide



Boat name	:	NM2-100
Last Saved	:	26 February 2025
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U-BOAT WORX B.V.



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Critical Piloting Numbers		
Author: Gido Grooten	Date: 17/08/2021	Version: 3
Checked: L. van Rijswijk	UBW ref: NM246D-11-02-07	Status: Final

REVISION TABLE			
Version	Date	Modifications	Name
0	18-05-2022	Initial Version	L. van Rijswijk
1	19-09-2022	Added skid support area	G. Grooten
2	02-05-2023	Added Cayman equipment, Added Slim Chargers, Added Battery life, Corrected Typo's, added optional thrusters, Added topside tablet, Emergency battery to 14Ah.	G. Grooten
3	05-02-2025	Added Weight optional thrusters	G. Grooten

DISTRIBUTION				
Organization	Name	Date	Version	Remarks



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1 Introduction

The specifications mentioned in this document apply to the NM2-100 submersible. This document is divided into a mechanical and an electrical section.

Despite careful editing it is not possible to completely rule out errors, misprints, or changes to third party products specifications. When in doubt, contact U-boat Worx.

2 General Layout

The values stated below are based on design calculations and can differ due to production tolerances and changes in equipment lay-out. The final measured values are noted in the DNV Appendix to Classification Certificate, which is located in the Certificates section of this logbook.

Dimensions are derived from NM2.46-000-0000-0 (Complete SUB Package), which is found in the System Drawing section of this logbook

Weight in Air (No Options)	2.500 kg
Dive ready, full payload	2.700 kg
Displacement	2.600 L
Occupancy	1 passenger and 1 pilot

Dimensions

(A) Length	2.800 mm
(B) Width	2.320 mm
(C) Height	1.550 mm
(D) Height hatch open	2.155 mm
(E) Freeboard	690 mm
(F) Skid Length	1.300 mm
(G) Skid Width	1.720 mm

Occupancy

No. of occupants	1 passenger and 1 pilot
Payload (No Options)	200kg
Max Payload*	250kg
	Payload is the sum of:
	- Passenger weight
	- Equipment weight
	- Trim weight

Drop Weight

Drop weight	100 kg
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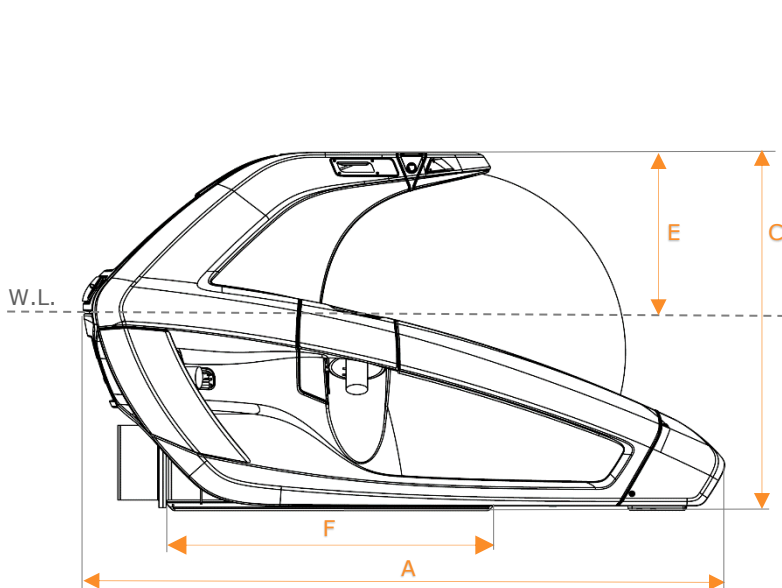


Figure 1: SB view floating configuration

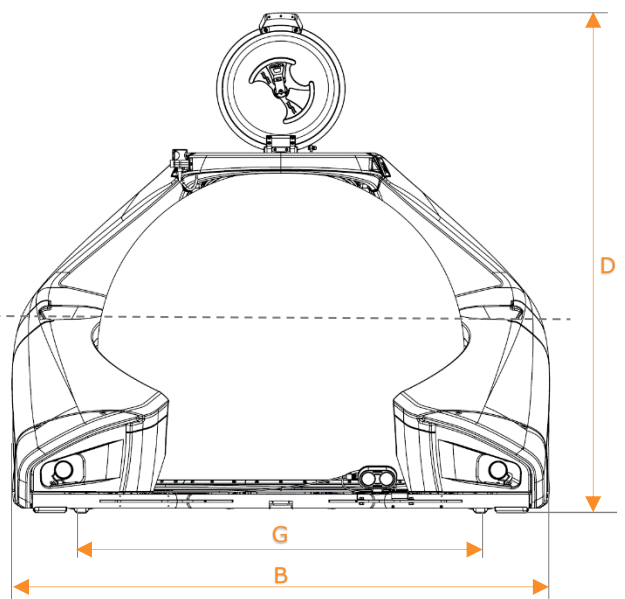


Figure 2: Bow view storage configuration

3 Operational Limits

3.1 General

Depth

Nominal Diving depth (NDP) 100 msw

Inclined position

In any direction max 22.5°
Transient inclination max 45°

Acceleration

Down max 2 G
Up max 1 G
Transverse max 1 G, in any direction

Environmental

Seawater density 1.028 kg/m³
Seawater salt content 3,5 %
Sea state max 2
Surface temperature - 2 °C to 32 °C
NDD temperature max 32 °C

Operating speed (calculated)

Surface longitudinal max 1,65 knots
Surface lateral max 1,33 knots
Submerged longitudinal max 1,32 knots
Submerged lateral max 1,06 knots
Submerged Vertical max 0,95 knots

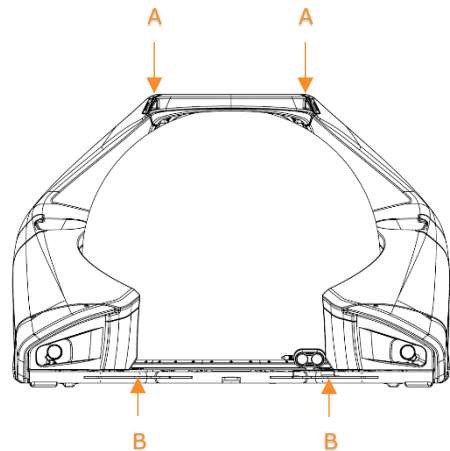


Figure 3: Lifting & Lashing Points

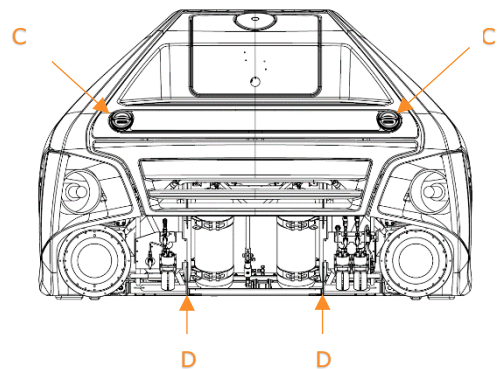


Figure 4: Mooring points & towing points

3.2 Launch & Recovery

PVHO lifting points

Lifting type Dual lifting point
Location Point A, Figure 3
Lifting point compatibility Round Sling or optional Lifting Beam
Lifting point distance 726 mm
Lifting point height from floor 1.550 mm
SWL per point vertical 2 t
Sling length $L \geq 750$ mm
Angle between lines $\alpha \leq 55^\circ$, Figure 5
SWL per point angled 2.45 t ($\alpha \leq 55^\circ$)

3.3 Towing

Limits

Towing speed max 3 knots
Sea state max 3

Towing points

No. of towing points 2 points
Location Points D, Figure 4
Towing direction longitudinal, stern

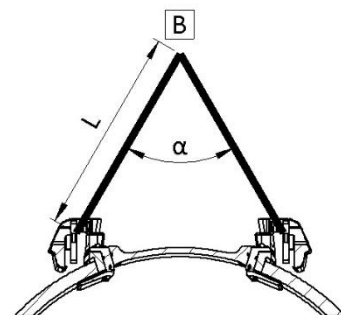


Figure 5: Sling Lifting

3.4 Storage and Transport

Limits

Temperature	-10 °C to 45°C
Environment	Submersible protected against: - Dust, - (salt)water spray, - direct sunlight and UV-radiation

Lashing points

No. of lashing points	4
Location	Points B + D , figure 3/4
SWL per point (A)	max 0,4 t in any direction
SWL per point (D)	max 0,4 t in any direction

Painter points

No. of painter points	2
SWL per point. Max	0,1t horizontal
Location	Points C , Figure 4

Skid Support Area

When placing the Nemo Submersible on support points like a custom framework or trailer, make sure the frame is primarily supported at support area **E1** to/and **E2**. Point **F** is secondary and not intended as a primary support area. Do NOT only support on **E1** and **F** without supporting **E2**.

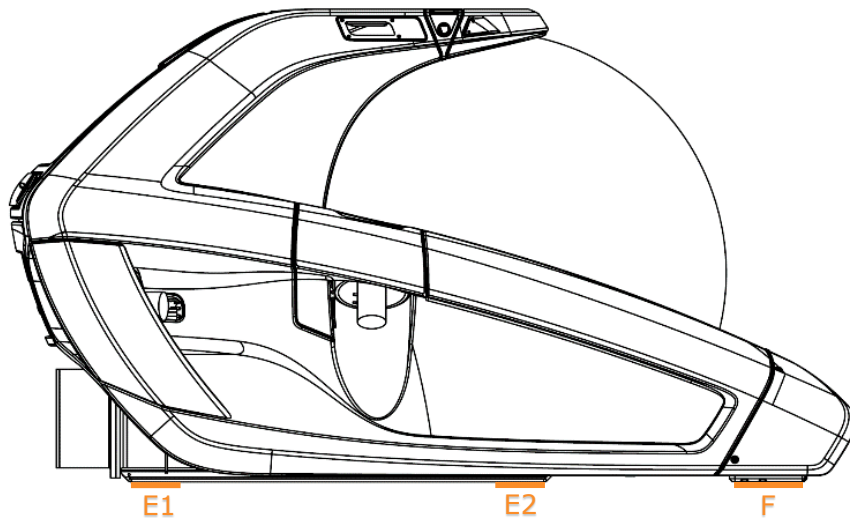


Figure 6: Primary and secondary support points

4 Protection systems

4.1 Dead Man Switch

The dead man switch (DMS) is implemented to make sure that the vessel always surfaces, even if the operator is unconscious.

Activation

The timer will start counting when these conditions are met:

- Depth sensor value should be valid
- Depth > 5msw

Reset

The timer will be reset when one of the pre conditions are omitted, or

- The DMS switch of the MANTA controller is pressed
- The DMS switch on the UIS screen is pressed

Sequence

$t \geq 00:09:00$

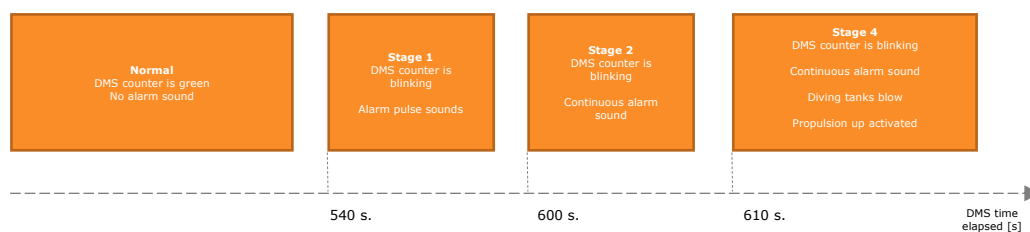
$t \geq 00:10:00$

$t \geq 00:10:10$

Stage 1: Blinking DMS button, pulsating beep

stage 2: Blinking DMS button, continuous alarm

stage 3: Blinking DMS button, continuous alarm, DT blow (1 sec every 5 sec), upward propulsion



4.2 Maximum Depth Protection

To ensure that the vessel cannot go deeper than the specified maximum depth of 100 msw, maximum depth protection is implemented.

Sequence

Depth > max. depth – 5 m

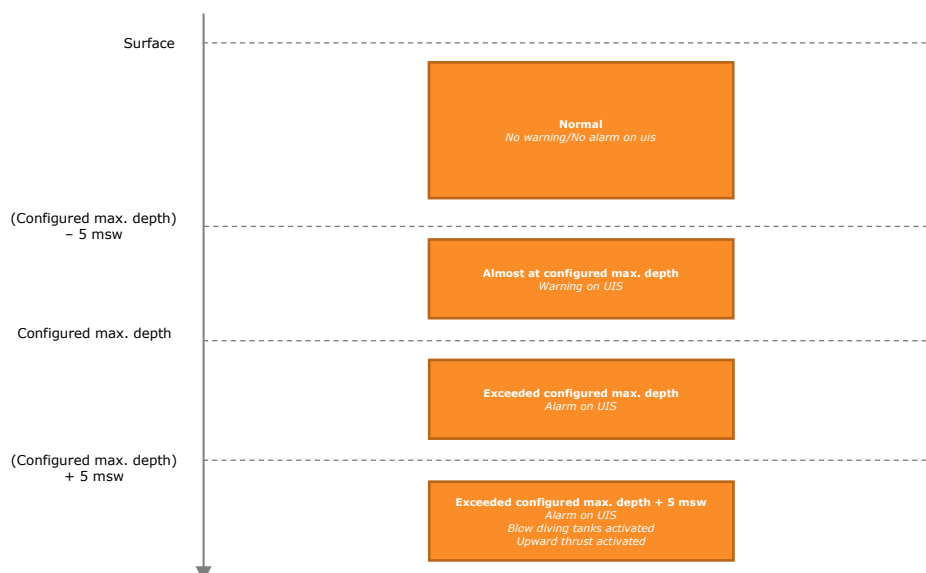
Depth $\geq$ max. depth

Depth $\geq$ max. depth + 5 m

UIS warning

UIS alarm and audible alarm buzzer

UIS alarm, Diving Tanks blow an upwards thrust activation until diving depth $\leq$ max. depth + 5 m



5 Systems

5.1 Pressure Hull

Type	Dry, atmospheric pressure
Hatch internal diameter	500 mm
Window design temperature	32 °C, ASME PVHO-1-2016 Section 2 2-2.4.1 (1)

External pressure

Nominal Diving Pressure (NDP)	10,1 bar
Test Diving Pressure (TDP)	14,14 bar

Internal pressure

Design pressure	800 – 1345 mbar
Pressure relief valve setting	Δ 69 mbar (1 psi)

Internal volume

Empty PVHO, no payload	1.410 L
------------------------	---------

5.2 Pneumatic System

Medium	Filtered air
No. of PV's	2
PV volume	15 L
Total volume	30 L
Working pressure	200 bar
Reserve pressure	min 100 bar

5.3 Aircon System

The limitations specified hereunder are based on an operating Aircon system.
The Aircon may only be operated with the heat exchanger.

Aircon Unit

No. of units	1
--------------	---

Environmental conditions

Internal temperature	10 °C to 41 °C, aircon limits
Internal humidity	max 90 %
External temperature	max 32 °C, heat exchanger limit

Coolant agent

Medium	R-134a
Fill weight	200 grams
Standby pressure	3,5 to 5,5 bar @ 20°C

Coolant fluid

Medium	Kroon Coolant SP 12
UBW art. no.	KROO002
Fluid colour	Pink
System volume	3 L
Aircon reservoir level	Between min and max



5.4 Hydraulic Systems

Hydraulic Oil

Medium	Panolin Atlantis 32
UBW art. no.	UBWT937
Medium type	Synthetic biodegradable hydraulic fluid
Interchangeable with	Panolin HLP SYNTH 32

Hydraulic system

Working pressure	max	18,1 bar
Compensating pressure		0,2-0,5 bar above ambient pressure
Compensator level		Between MIN and MAX
System oil volume		0,8L
Functions		- Vent Valves

Vent valves

Opening time	max	NA
Closing time spring return	max	3 sec

Hydraulic emergency system

Working pressure	max	60 bar
Compensating pressure		0,2-0,5 bar above ambient pressure
Compensator level		Between MIN and MAX
System oil volume		0,9 L
Functions		- Drop Weight Release - Buoy Release - Manipulator Release

5.5 Buoyancy System

Bilge Pump

Type		Impeller pump
UBW art. No.		NEMA002
Pump capacity	min	15 L/min
Self-priming		3,0 m
Operating pressure	max	0,6 bar
Power consumption		100 W

Bilge inline filter mesh

Type	in-line filter
Filter UBW art. no.	ubwj756

5.6 Diving Tanks

Type	GRP tank with vent valve on top
No. of tanks	4
Tank volume Bow	332 L
Tank volume Stern	174 L
Total volume	$2 \times 332 \text{ L} + 2 \times 174 \text{ L} = 1.012 \text{ L}$
Required pressure full blow	33,7 bar



5.7 Propulsion System

Longitudinal, lateral & vertical manoeuvring

Thruster type		T1083 / T1084	
UBW art. no.		NEMA962 / NEMB609	
Weight.		11.8Kg	
No. of thrusters		2/4	
Input voltage	nominal	130 VDC	
Power consumption	max	2.4 kW	
Forward bollard trust	max	480 N	(105 Lbs)
Reverse bollard trust	max	480 N	(105 Lbs)
Depth rating	max	850 msw	

(Optional) Longitudinal manoeuvring

Thruster type		TZ2041S	
UBW art. no.		NEMB181	
Weight		20Kg	
No. of thrusters		2	
Input voltage	nominal	130 VDC	
Power consumption	max	4.5 kW	
Forward bollard trust	max	900 N	(202 Lbs)
Reverse bollard trust	max	700 N	(157 Lbs)
Depth rating	max	850 msw	

6 Life Support

6.1 Oxygen Supply

O₂ grade

Grade	Medical / Grade 2.5
Purity	≥ 99,5 %
CO	< 5 ppm
CO ₂	< 300 ppm
H ₂ O	< 67 ppm

O₂ Main Supply

No. of PV's	1
PV volume	2 L
Working pressure	max 232 bar

O₂ Min Operating time

O ₂ Consumption	26 L/h per person
Total O ₂ volume	2 x 200 = 400 L
Operating time	max 15,4 man-hours
2 occupants	max 7,7 hours

O₂ Reserve Supply

No. of PV's	2
Individual PV volume	15 L
Total volume	30 L
Working pressure	max 200 bar
Reserve pressure	min 170 bar
Operating time	> 96 h

6.2 CO₂ Removal System

CO₂ Scrubber

Type	Radial flow CO ₂ scrubber
Absorbent capacity	2,5 kg
Canister mounting	Toggle Clamp locking mechanism
Canister mesh size	0,3 mm
Fan power consumption	24 W

CO₂ Absorbent

Medium	Sofnolime CD Grade
UBW art. no.	LIES064
Particle size	2,0 to 5,0 mm
Cross-section	D-shaped
CO ₂ absorption capacity	max 110 L CO ₂ /kg
Storage temperature	0 °C to 35 °C

CO₂ scrubber Operating Time

CO ₂ Production	22 L/h per person
Total removal capacity	2,5 x 110 = 275 L CO ₂
Operating time	max 12,5 man-hours
2 occupants	max 6,25 hours



6.3 Air Monitors

Type	Dräger X-am 5600		
No. of air monitor	2, completely independent		
Sensors	O ₂ , CO ₂ and H ₂		
UBW art. no.	DAAN041		
<u>O₂ sensor</u>			
Concentration		19 % ≤ O ₂ ≤ 23,5 %	(Alarm)
Fresh air calibration		every 6 months	
Span calibration		every 12 months	
<u>CO₂ sensor</u>			
CO ₂ Concentration	max	1,0 %, normal operation	(Alarm)
	max	2,0 %, during emergencies.	(Alarm)
Expected sensor life		> 5 years	
Fresh air calibration		every 6 months	
Span calibration		every 12 months	
<u>H₂ sensor</u>			
H ₂ concentration	max	1.500 ppm	(Alarm)
Span calibration		every 12 months	

6.4 Freon Sensor

Type	BUTOX ST350EX		
UBW art. no.	UBWM861		
Span calibration	Every 12 months		
R-134a concentration	warning	750 ppm	(Alarm)
	max	1.000 ppm	(Alarm)

6.5 Fire Extinguishing

Type	Zero 360 1.0kg Novec 1230 handheld extinguisher		
UBW art. no.	NEMA546		
Activation system	Manual (dual action)		
Extinguishing agent	Novec 1230		
Agent per container	1 kg		
Cylinder filling pressure	7 bar		
Mass	1.58 kg		
Operating temperature	-15 °C to 60 °C		
Fire Class Suitability	A - Solids		
	B - Liquids		
	C - Gasses		

6.6 Emergency Food

+6Type	Seven Oceans Emergency Food Ration 500 g		
UBW art. no.	UBWG126		
Pack content	18 tablets packed in 9 bars		
Energy	Total 10.000 kJ / 2.500 kcal per pack		
Shelf life	5 years		
Consumption rate	3.333 kJ per person per day		
Total packs	3		

6.7 Emergency Water

Type	Seven Oceans Emergency Drinking Water		
UBW art. no.	UBWG129		
Pack content	0,5 L water		
Shelf life	5 years		
Consumption rate	0,5 L per person per day		
Total amount	8 packs		



6.8 CO₂ Scrubber Curtains

Type	Lithium Hydroxide Curtains
UBW art. no.	UBWG115
Pack content	4 LiOH curtains
Absorbent Performance	132 L CO ₂ per sheet
Shelf life	8 years
Total amount	9 packs

6.9 Oxygen Self Rescuer

Type	Oxy K 30 HS
UBW art. no.	UBWG133
Breathing duration	30 min at 35 L/min 120 min at 10 L/min
Shelf life	10 years
Total amount	2 units + 1 spare

6.10 Life Jacket

Type	EAM UXF-35 Lifejacket
UBW art. no.	UBWQ476
Activation system	Manual activation
Additional equipment	Whistle and water activated light
Total amount	2 jackets

6.11 Cayman Equipment

First Aid Kit

Type	First Aid Kit CAT C
UBW Art No.	UBWB589
Shelf Life	5 Years

Handheld Compass

Type	IRIS 50
UBW Art No.	UBWE455

Digital Stopwatch

Type	RS PRO
UBW Art No.	BEEK025



7 Battery specifications

7.1 External Battery Packs

Battery type	Lithium Iron Phosphate Battery (LFP)
Number of packs	4
Total power capacity	17,9 kWh
Operating temperature	-29 °C to 59°C

Charging

Charge time	nominal	4 hours
	max	6 hours
Charging temperature		-9 °C to 50 °C

State of Charge (SoC)

Operational level	min	20 %
Absolute minimum		0 %
Storage level		40 %
Storage time	≥ 25 °C	2 months
	≥ 40 °C	< 1 week

7.1.1 24V Pack 1 & 2

Battery type	24 VDC LFP pack
No. of cells per 24V pack	16 cells
No. of 24V packs	2
Total power capacity	2 x 2,56 kWh = 5,12 kWh
Voltage	min 20
	max 28,8 V
Cell V difference	target 10 mV
Insulation monitor	> 2 kΩ
Powered systems	- Interior & life support systems - Automation, Communication Navigation & Locating devices - 24V External lights

7.1.2 130V Pack 1 & 2

Battery type	130 VDC LFP pack
No. of cells per 130V pack	40 cells
No. of 130V packs	2
Power capacity	2 x 6,4 kWh = 12,8 kWh
Voltage	min 100 V
	max 144 V
Cell V difference	target 10 mV
Insulation monitor	min > 20 kΩ
Powered systems	- Thrusters - 130V External flood lights

7.2 Emergency Internal Battery Pack

Type battery	VRLA Gel battery
UBW art. no.	NEMB101
No. of batteries	2
Power capacity	14 Ah
Voltage	min 22,0 V
	nominal 24,0 V
	max 27,5 V
State of charge (Soc):	SoC cannot be determined on battery voltage
Storage time	≤ 40 °C 3 months
Powered systems	96 H life support systems - Internal light - Switch led - UWT - Air Monitor Back-up



7.3 Battery Chargers

7.3.1 Battery Charging Station

Power input requirements 3-Phase

Voltage	3 Phase 400Vac +N +PE
Current	16 A
Frequency	50 to 60 Hz
Connector	CEE CST 5pin/16A/400V

Power input requirements 3x Single Phase

Voltage	3 x 110-240 Vac ¹
Current	3 x 16 A
Frequency	50 to 60 Hz
Connector	3x Mains Plug CEE 7/7 DE Schuko / F, 16A, 250V

7.3.2 24V Battery Charger (Optional)

Charger type	Delta Elektronika SM 66-AR-110		
No. of chargers	1		
<u>Power Output</u>		<u>Normal</u>	<u>Trickle</u>
Output voltage	max	30,0 VDC	26,2 VDC
Output current	max	54,5 A	54,5 A

7.3.3 130V Battery Charger (Optional)

Charger type:	Delta Elektronika 330-AR-22		
No. of chargers	1		
<u>Power Output</u>		<u>Normal</u>	<u>Trickle</u>
Output voltage	max	138,6 VDC	128,5 VDC
Output current	max	22 A	22 A

7.3.4 24V Battery Charger (Slim)

Charger type	Mean Well NPB-1700-24		
No. of chargers	1		
<u>Power Output</u>		<u>Normal</u>	
Output voltage	max	29,1 VDC	
Output current	max	50 A	

7.3.5 130V Battery Charger (Slim)

Charger type:	Mean Well NPB-1200-48		
No. of chargers	2		
<u>Power Output</u>		<u>Normal</u>	
Output voltage	max	136,8 VDC	
Output current	max	18 A	

7.3.6 Emergency Battery Charger

Charger type	Phoenix Contact QUINT-PS/24DC/24DC/ 5		
UBW art. no.	NEMA035		
No. of chargers	1		
Output voltage	max	27.9 VDC	
The emergency battery is charged when the 24V main charging plug is connected.			
The emergency battery is full at 27.5V.			

¹ Applying a power source less than 200Vac to Battery Charger (Slim) will reduce charging speed.

8 Automation, Communication, Navigation & Locating Devices

8.1 Underwater Telephone (UWT)

UWT Unit

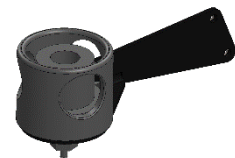
Type	STX OEM PCB
UBW art. no.	UBWU927
Operating frequencies	25 kHz
Transmitter activation	PTT (push-to talk)
Transmitter power	70 Watts
Range	max 6 km
Supply voltage	24 VDC

Headset

Type	BRH441MLC Headset w/ Boom Mic
UBW art. no.	UBWM065

Transducer

Type	Transducer for Bell-200 system, 25kHz
UBW art. no.	UBWA179
Beam pattern	Hemispherical
No. of transducers	1
Location	Hatch tunnel



8.2 VHF

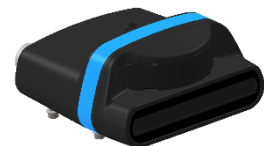
Handheld VHF

Type	ICOM VHF IC-M94DE
UBW art. No.	UBWJ521
Frequency coverage	TX: 156.000 - 161.450 MHz RX: 156.000 - 163.425 MHz
IP rating	IPX7, with battery pack attached
Battery life	approx. 9 hours
Charging time	approx. 3 hours



8.3 Sonar

Type	Oculus M750d Multibeam Sonar System
UBW art. no.	NEMA286
Operating frequency	750 kHz / 1.2 MHz
Field-of-view	130° x 20° / 70° x 12°
Maximum range	120 m / 40 m
Minimum range	0,1 m
Range resolution	4,0 mm / 2,5 mm
Update rate	40 Hz
Connectivity	Ethernet
Supply voltage	18 to 32 VDC
Current drain	max 10 W at 750kHz max 35 W at 1.2MHz
Depth rating	500 msw
Housing material	Anodised Aluminium



8.4 USBL

Type		SeaTrac X150
UBW art. no.		NEMA285
Data interface		RS232
Operating range	max	1000 msw
Frequency band		24 to 32 kHz
Supply voltage		9V to 28V VDC
Operating temperature		-5 °C to 40 °C
Depth rating	max	2000 msw
Housing materials		316 Stainless steel

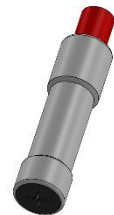
Power consumption

Stand-by mode	Approx 0.6 W
Transmit mode	Approx 6 W



8.5 Altimeter

Type		Valeport VA500
UBW art. no.		NEMA303
Operating frequency		500 kHz
Conical beam width		±3°
Range	min	0.1 msw
	max	100 msw
Supply voltage		9 – 28V DC
Current drain		125 mA @ 12V
Depth rating	max	6000 msw
Housing materials		Titanium



8.6 Lights

24V Spot light

Type		24V Spot LED Light 6.000LM
UBW art. no.		NEMA682
No. of spot lights standard		2
No. of spot lights additional option		2
Light output		6.000 Lumen
Beam angle		10°
Supply voltage		24 VDC
Power consumption	max	60 W
Depth rating	max	300 msw



260V Flood light

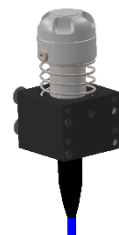
Type		260V LED Flood Light 52000LM
UBW art. no.		NEMA681
No. of flood lights standard		1
No. of flood lights additional option		2
Light output		52.000 Lumen
Beam angle		74°
Supply voltage		130/260 VDC
Power consumption	max	350 W
Operating depth	max	300 msw



8.7 Top Light

Top light

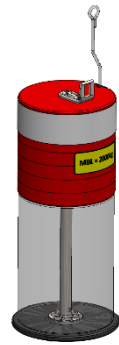
Beam angle	180°
Light output	350 Lumen
Supply voltage	8 - 26 VDC
Power consumption	5 W



8.8 Release Buoy

Buoy

UBW art. no.	NEMX116
Length	515 mm
Diameter	170 mm
Weight	max 7 kg
Weight without rope	2 kg
Floating position	upright
Floating height out of the water	135 mm
Rope type	Dyneema Pur PU 12
Rope length	150 m
Rope diameter	5 mm
Buoy colour	Red, with reflective markings
MBL	2 t



8.9 Manipulator Arm (Option)

Type	BRAVO 5
UBW art. no.	NEMA608
Degrees of freedom	4
Length of arm	750 mm
Arm rotation	359° not continuous
Jaw closing force	2 kN
Jaw rotation	360° continuous
Torque	38 Nm
Lift capacity (full reach)	12 kg
Lift capacity (max)	20 kg
Controller type	Master Arm
UBW art. no.	NEMA975





9 Operational Equipment

9.1 Topside Communication Box

UWT Unit

Type		STX-101SB
UBW art. no.		BEEK897
Operating frequencies		25 kHz
Transmitter activation		PTT (push-to talk)
Transmitter power		70 Watts rms
Range	max	6 km
Battery voltage		12 VDC
Supply voltage		100-240VAC
Battery life		approx. 6 hours

Handheld microphone

Type	STX-101SB Hand-Held Microphone
UBW art. no.	UBWB899

9.2 USBL

USBL Transceiver

Type		SeaTrac X150
UBW art. no.		NEMA285
Data interface		RS232
Operating range	max	1000 msw
Frequency band		24 to 32 kHz
Supply voltage		9VDC to 28VDC
Operating temperature		-5 °C to 40 °C
Depth rating	max	2000 msw
Housing materials		316 Stainless steel
Stand-by mode		Approx 0.6 W
Transmit mode		Approx 6 W

USBL Topside Box

UBW art. no.	NEMX513
Data interface	WIFI
Battery voltage	5VDC
Supply voltage	100-240VAC
Battery life	approx. 15 hours
Charging time	approx. 4 hours

USBL Topside Tablet

Type	Panasonic FZ-G2
UBW art. no.	NEMB197
Data interface	WIFI
Battery voltage	11.4VDC
Supply voltage	100-240VAC
Battery life	approx. 12 hours
Charging time	approx. 4 hours